Transformers in Finance: Leveraging Attention Mechanisms for Time Series Forecasting

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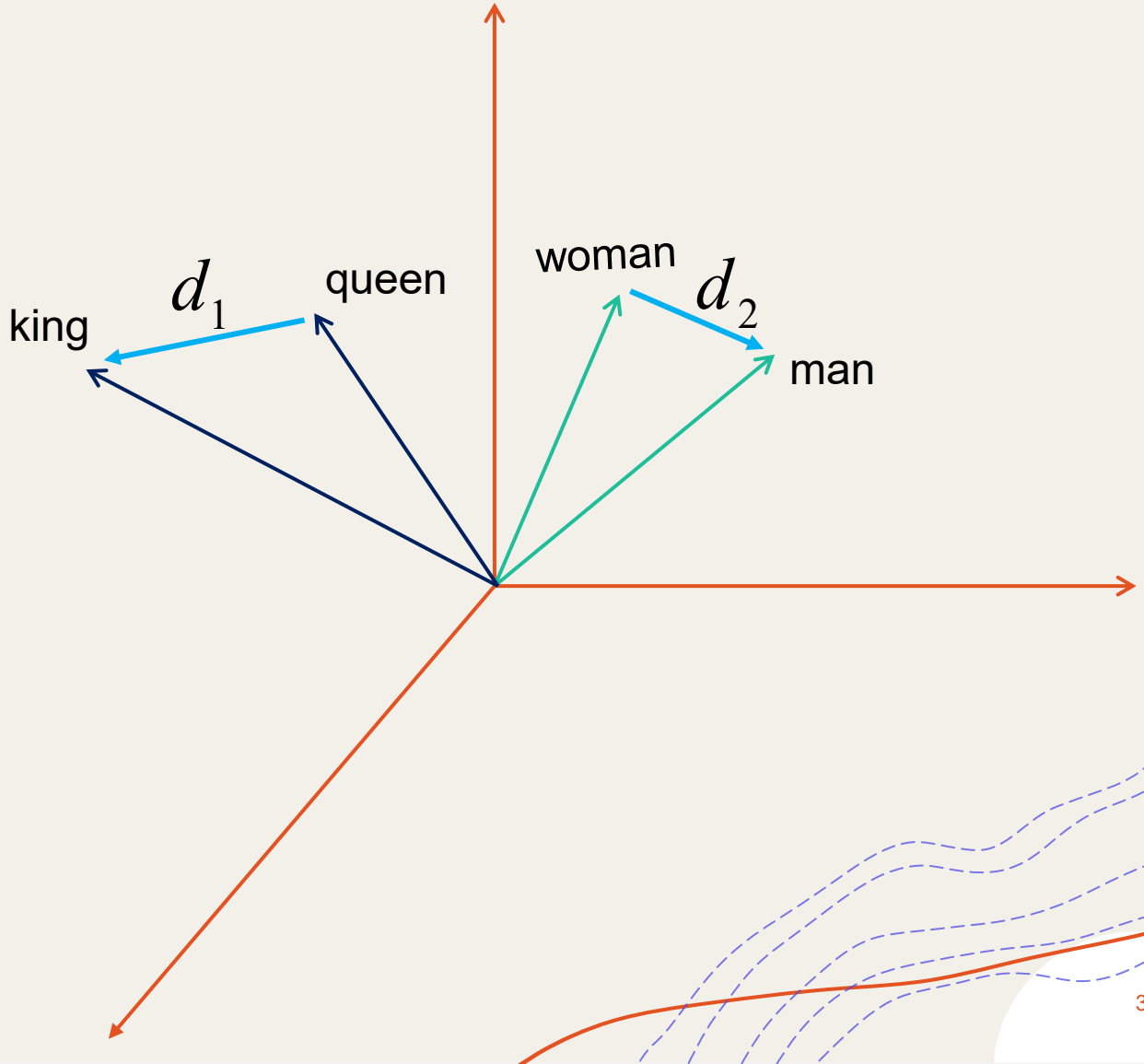
- May 2025

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- Types of Attention Mechanisms
- Examples of Self-Attention Mechanism in Time Series Analysis
- Transformers
- Application of Transformers in Time Series Analysis
- Strengths and Weaknesses of Transformers
- Hybrid Models Using Transformers

▪ **NLP*** and Attention Mechanism

- What is NLP?
- Word Embedding



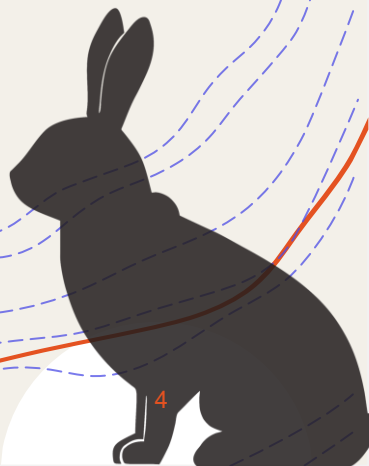
* Natural Language Processing

- NLP and **Attention Mechanism**

…… jumps over the fence.

- ❑ Bee
- ❑ Fox
- ❑ Rabbit
- ❑ Viper

……	**jumps**	over	the	fence



NLP and **Attention Mechanism**

- An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors.[1]

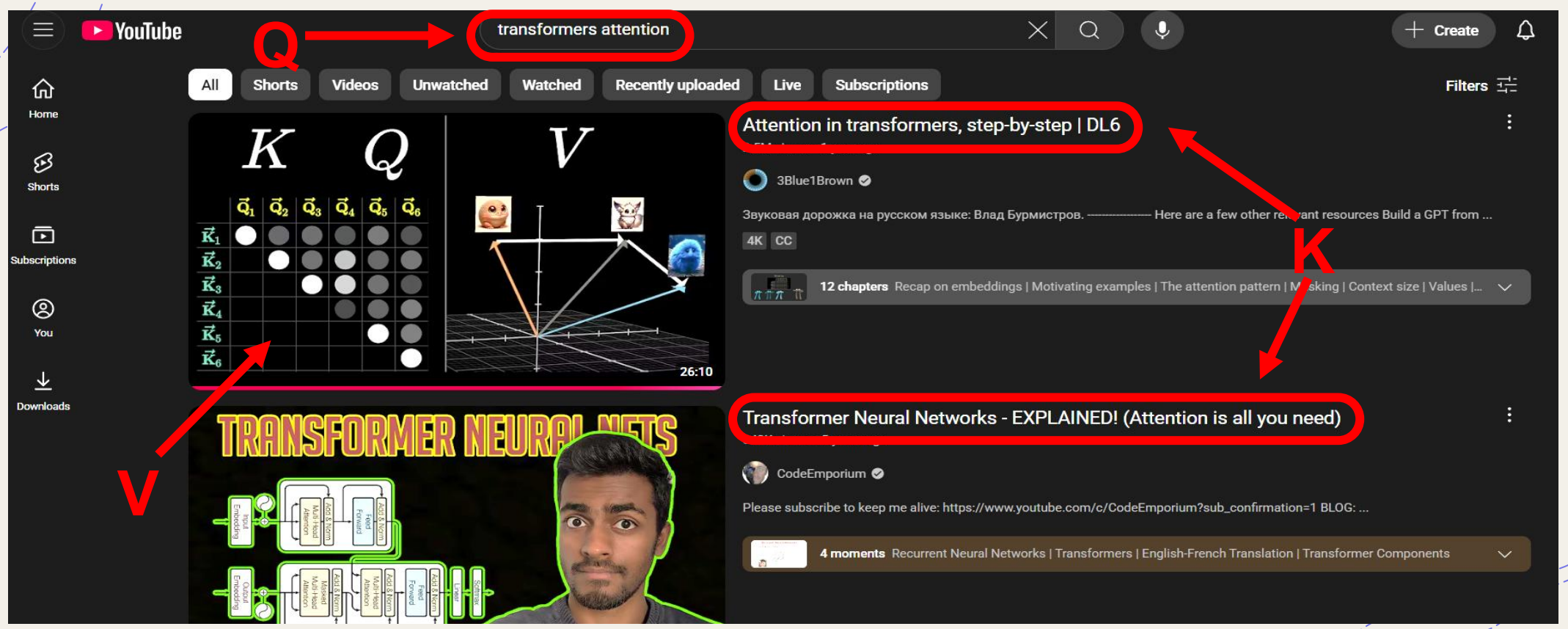


Fig 1 – Query, Key, and Values in youtube search

https://www.youtube.com/watch?v=eMlx5fFNoYc

- NLP and **Attention Mechanism**

$$PE_{(pos,2i)} = \sin(pos / 1000^{2i/d_{\text{model}}})$$

$$PE_{(pos,2i+1)} = \cos(pos / 1000^{2i/d_{\text{model}}})$$

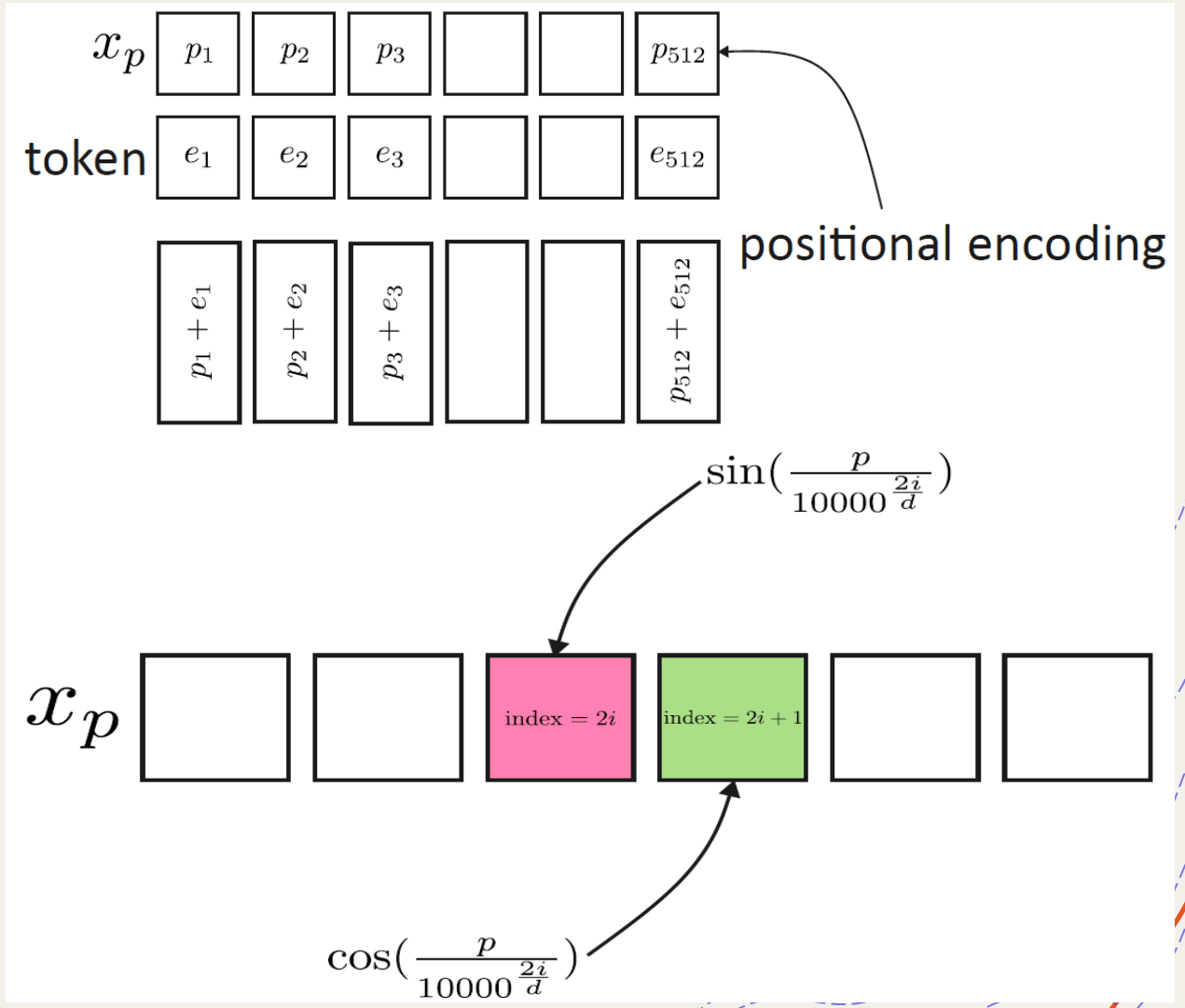
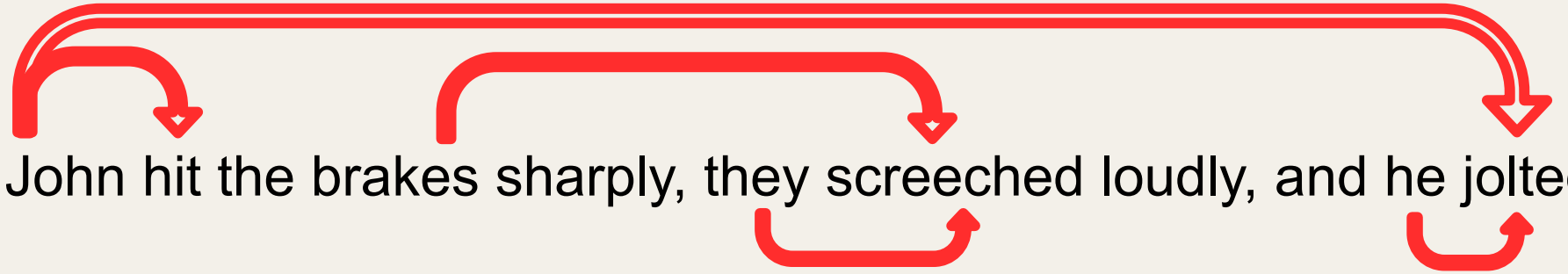


Fig 2 – Positional Encoding

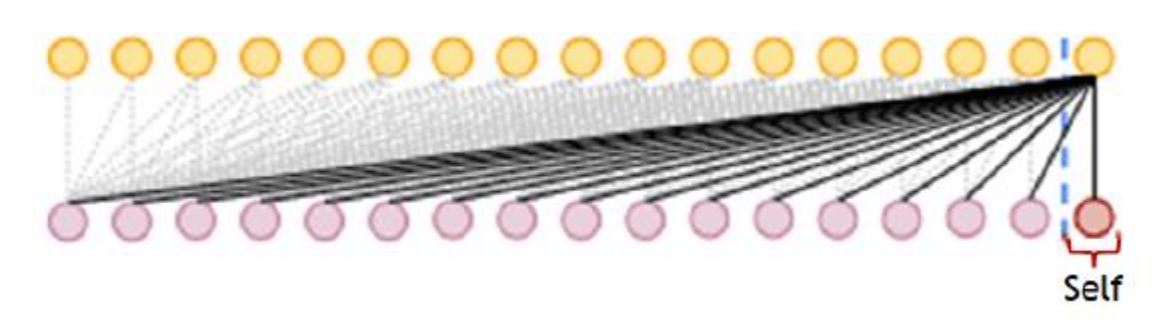
Types of Attention Mechanisms

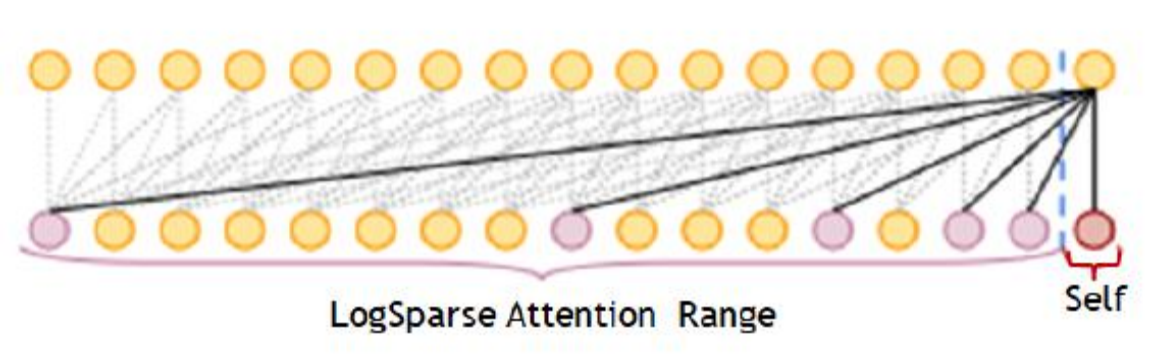
- Self-Attention
- Cross Attention

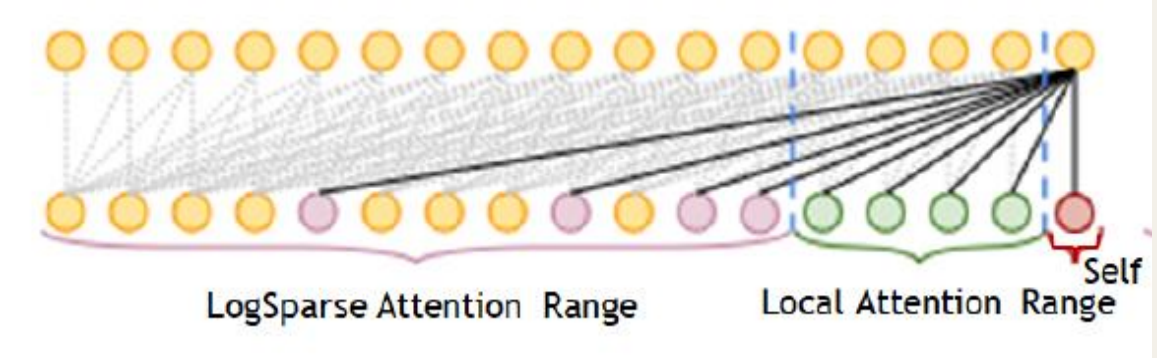


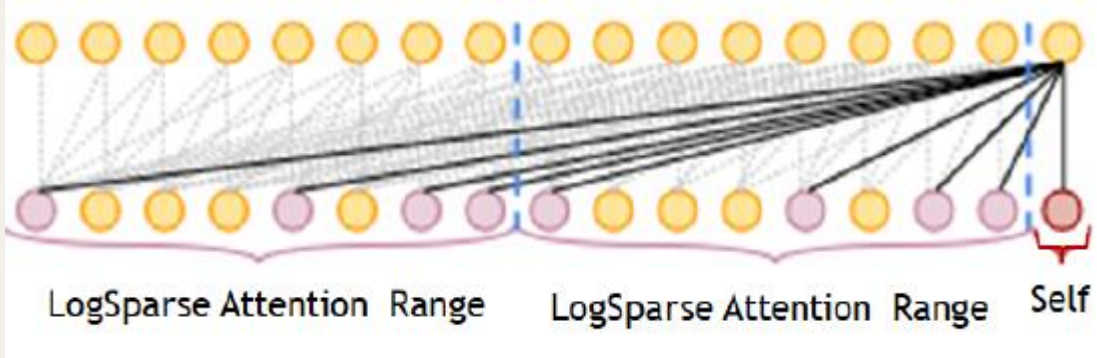


Examples of Self-Attention Mechanism in Time Series Analysis









Types of Self-Attention used in Time Series Analysis:

1. Full self-attention
2. LogSparse self-attention
3. Local Attention + LogSparse self-attention
4. Restart Attention + LogSparse self-attention

Fig 3 – four types of Self-Attention used in Time Series Analysis[6]

- Transformers



Attention Is All You Need

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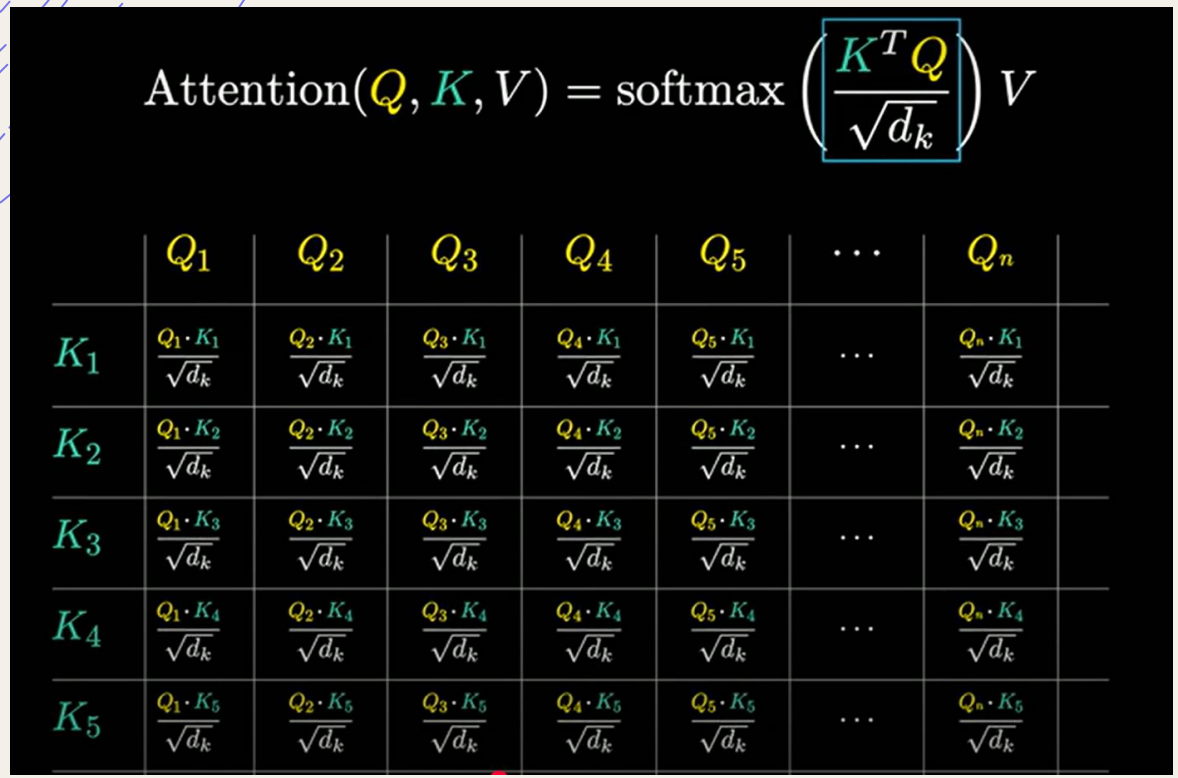
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Fig 4 – The first revolutionary article in this field[1]

- Transformers

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{K^T Q}{\sqrt{d_k}}\right) V$$



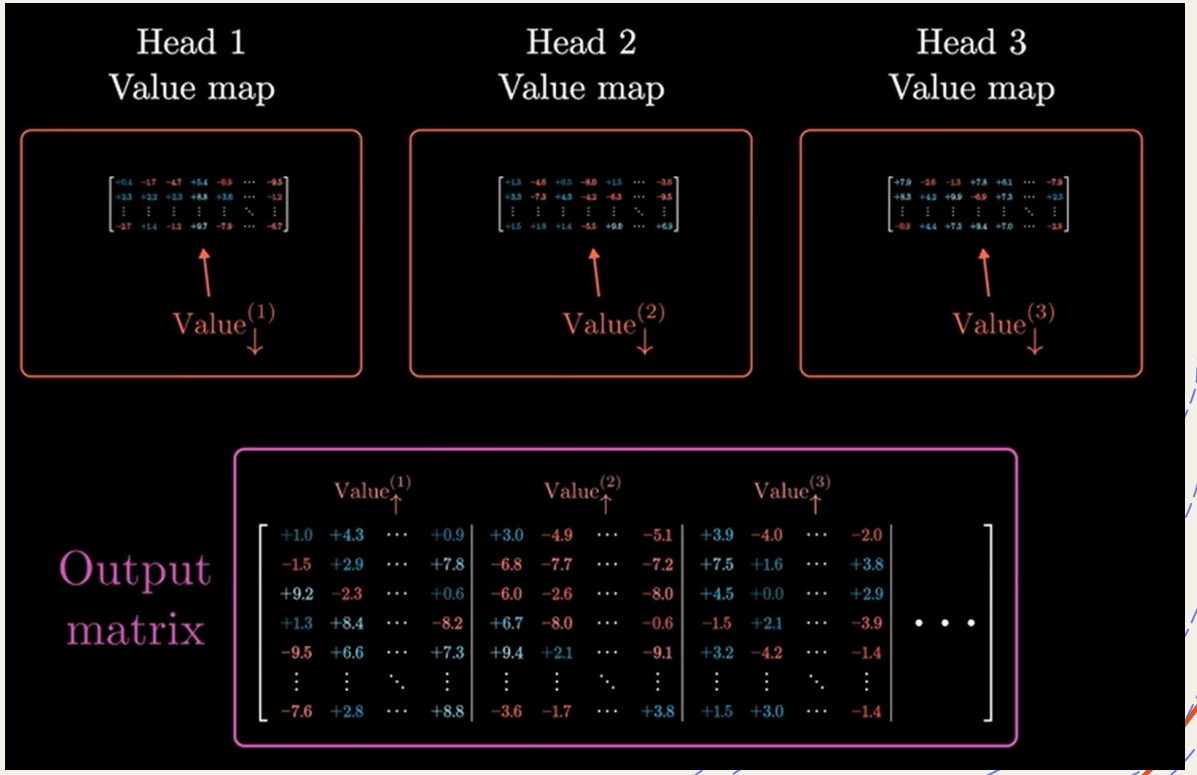


Fig 5, 6 – Output of Attention mechanism

- Transformers

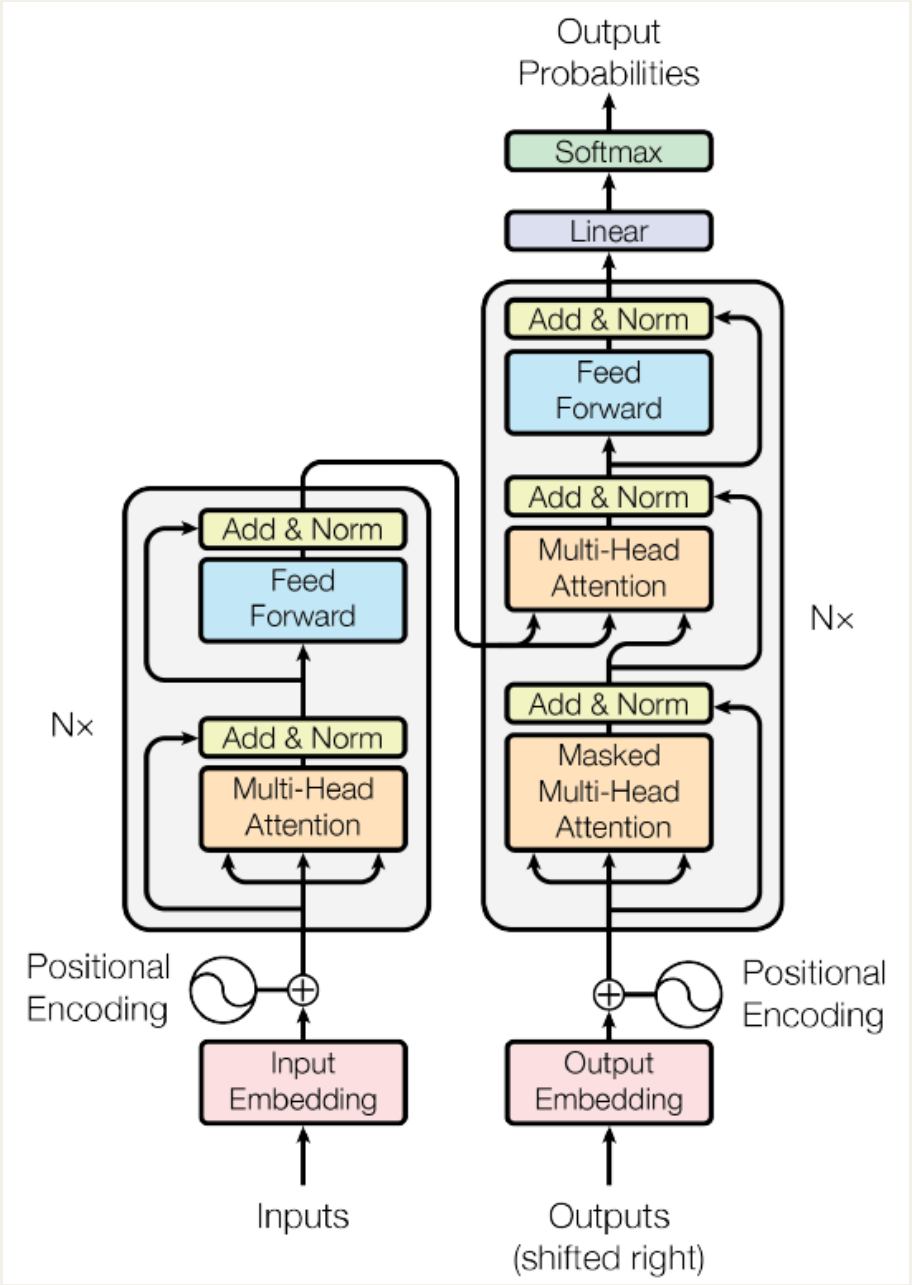


Fig 7 – Transformer Architecture[1]

Transformers

Type of Transformers:

- Standard Transformer
- **Masked Transformer**

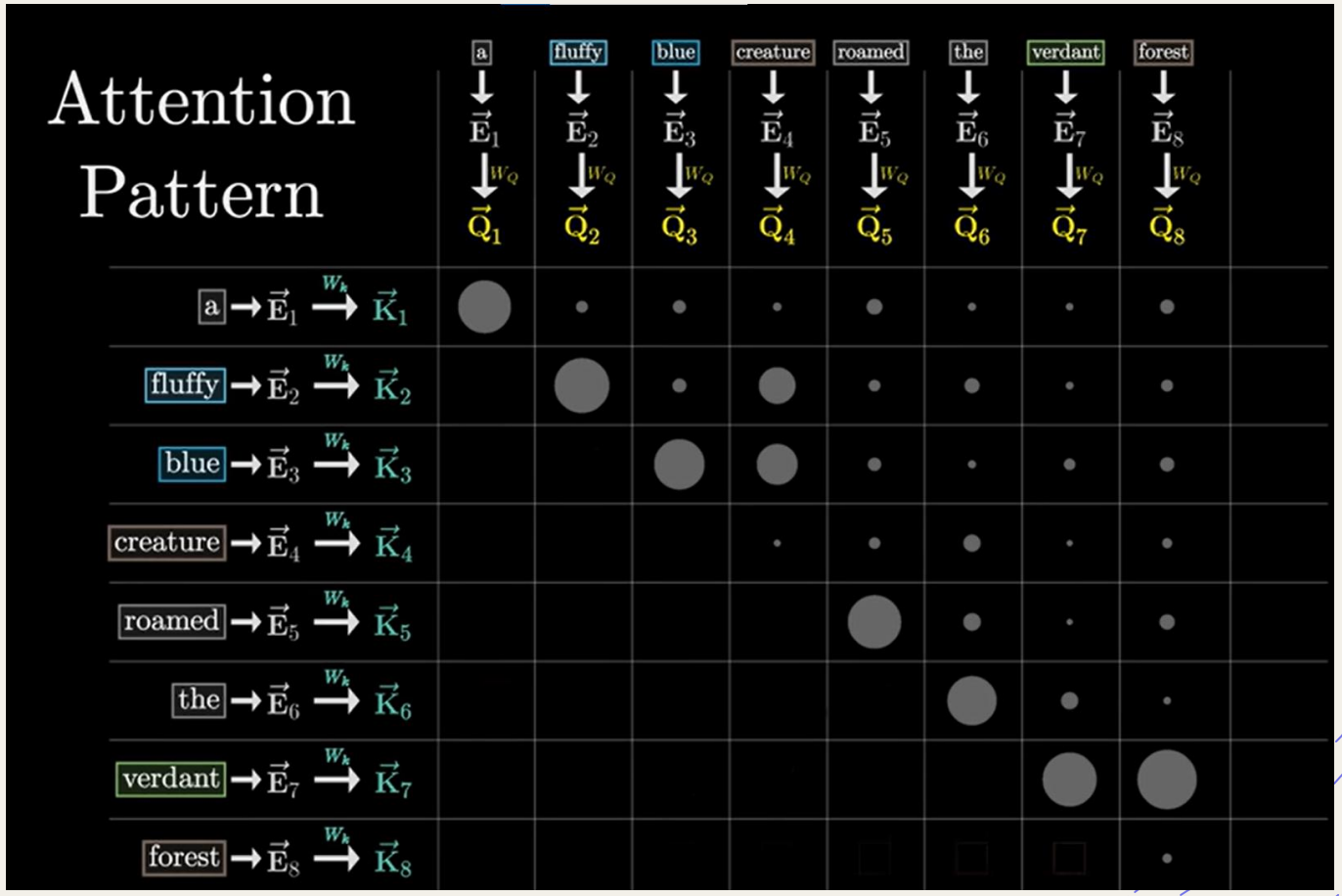


Fig 8 – Masked Transformer

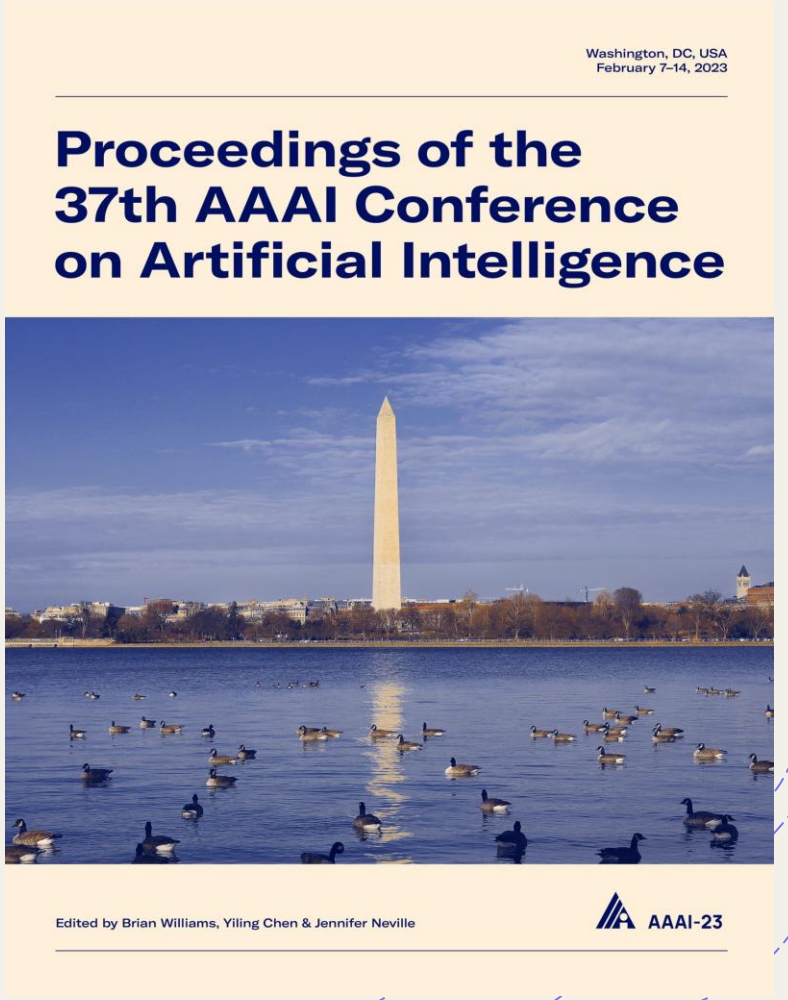
Application of Transformers in Time Series Analysis

Are Transformers Effective for Time Series Forecasting?

Ailing Zeng[1,2*], Muxi Chen[1*], Lei Zhang[2], Qiang Xu[1]

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Fig 9 – An important study about effectiveness of transformers for time series forecasting[2]



"we hypothesize that long-term forecasting depends on whether models can capture the **trend** and **periodicity** well."

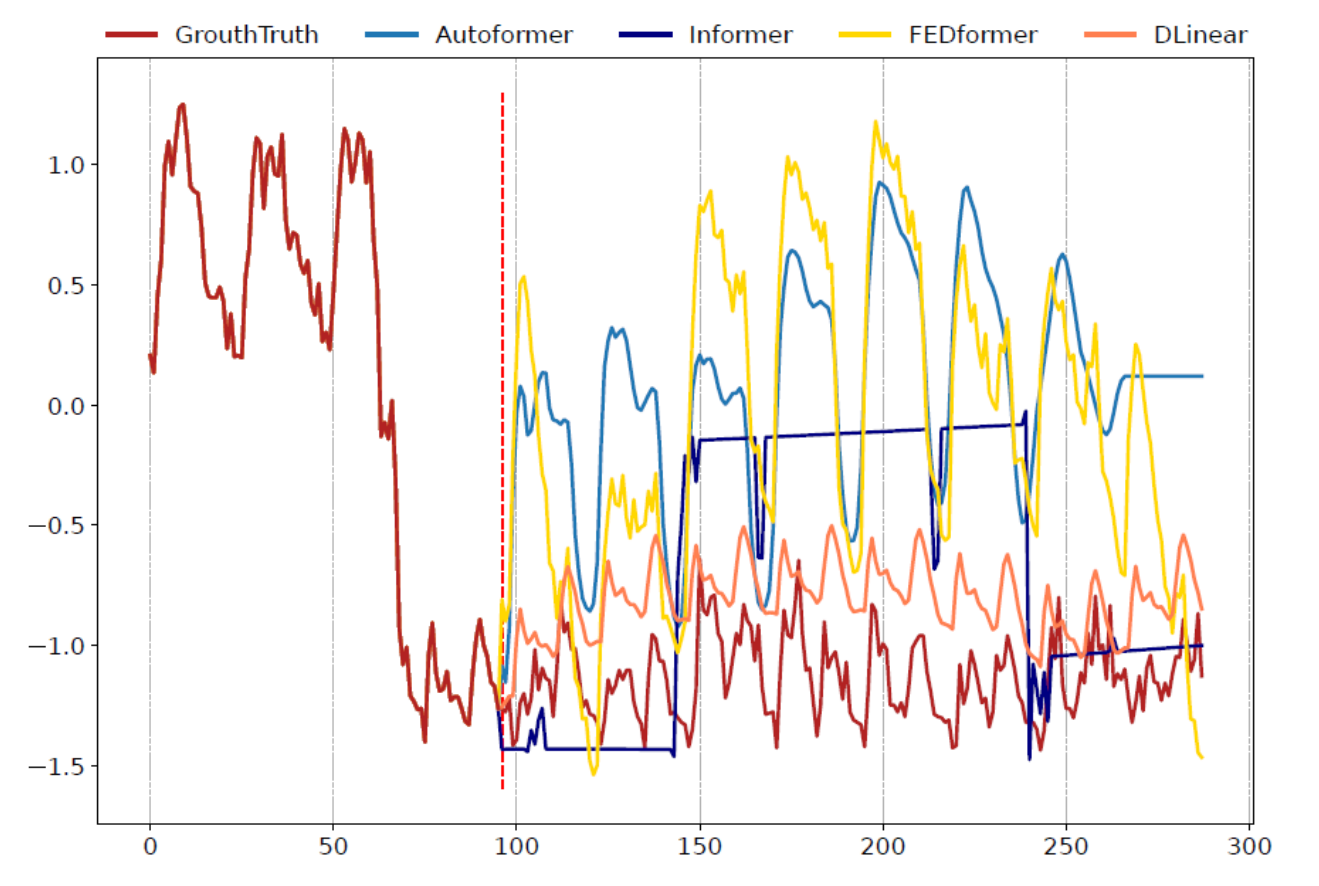


Fig 10 - comparing the result of 4 types of Transformers with the actual result[2]

Strengths and Weaknesses of Transformers

Strengths:

1. Parallel computation
2. Long-Term dependencies
3. Learning global context
4. The most successful sequence modeling architectures in NLP
5. Most successful solution to extract the **semantic correlations** among the elements in a long sequence

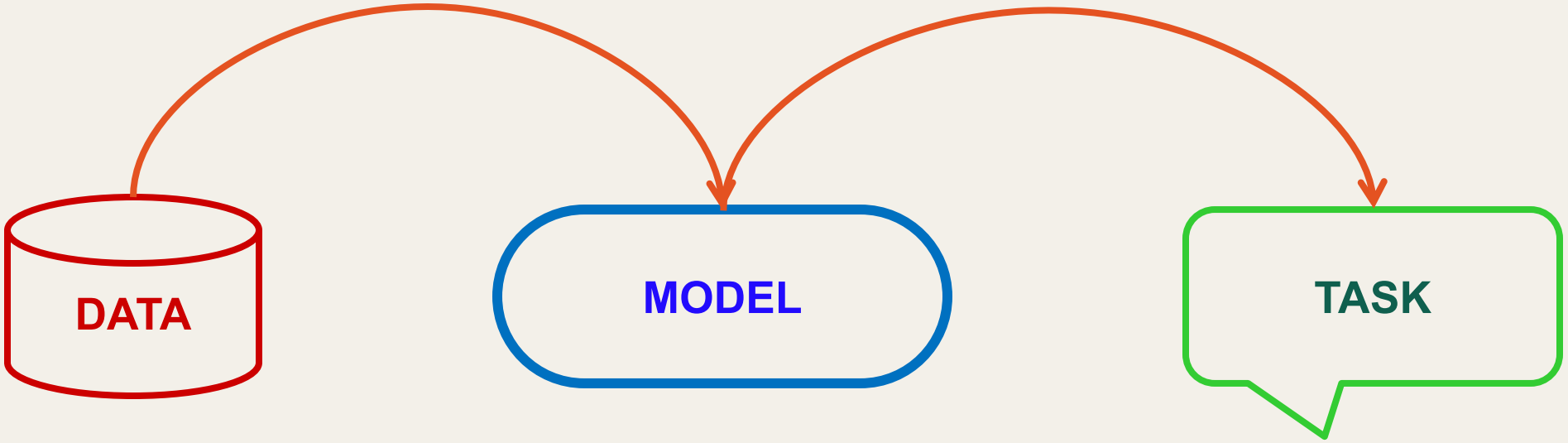
Strengths and Weaknesses of Transformers

Weaknesses[2]:

1. Permutation-invariant
2. Anti-Order
3. Fail to extract temporal relations
4. Fail to capture the scale and bias of the future data
5. Hardly predict the proper trend on aperiodic data
6. Can’t model any spatial correlation

Hybrid models using Transformers

Data-driven models and Extrapolating Patterns



Hybrid models using Transformers



Data-driven models and **Extrapolating Patterns**

- XGboost vs linear Regression
- CNN vs Transformers
- RNN vs Transformers

- Transformers aimed to overcome the limitations of recurrent neural networks (RNN) and convolutional neural networks (CNN) in **text processing** tasks. The initial goals of using Transformers in deep learning focused on **improving natural language processing**.

- Convolutional Neural Networks (CNN) are good at capturing **local patterns** for modeling **short-term** dependencies.[3]



Hybrid models using Transformers

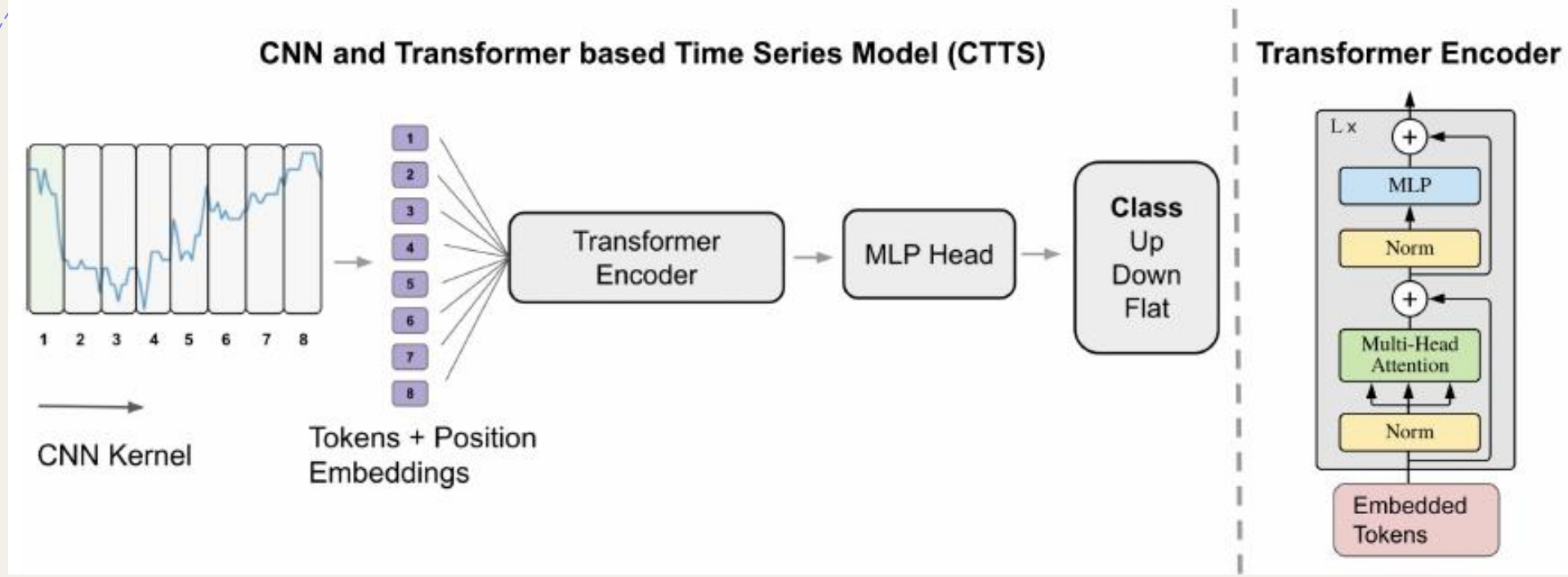


Fig 11 – CCTS architecture[3]

- For **CTTS** In our experiments, we used cross-entropy loss, and AdamW (Loshchilov and Hutter 2017) optimizer with batch size of 64 and max epoch of 100. we used CNN kernel size of 16 and stride 8. Transformer with depth 4 and 4 self-attention heads, with an embedding dimension of 128, and a drop rate of 0.3 to prevent overfitting.

- Hybrid models using Transformers

Method	2-class ↑	2-class* ↑	3-class ↑	3-class* ↑
EMA	53.2%	59.9%	39.5%	41.7%
ARIMA	50.9%	51.8%	37.5%	38.4%
DeepAR	51.1%	53.6%	37.4%	38.7%
CTTS	**56.7%**	**66.8%**	**44.1%**	**55.2%**

Table 1 – Summary of sign accuracy over the last quarter of 2019.[3,5]

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